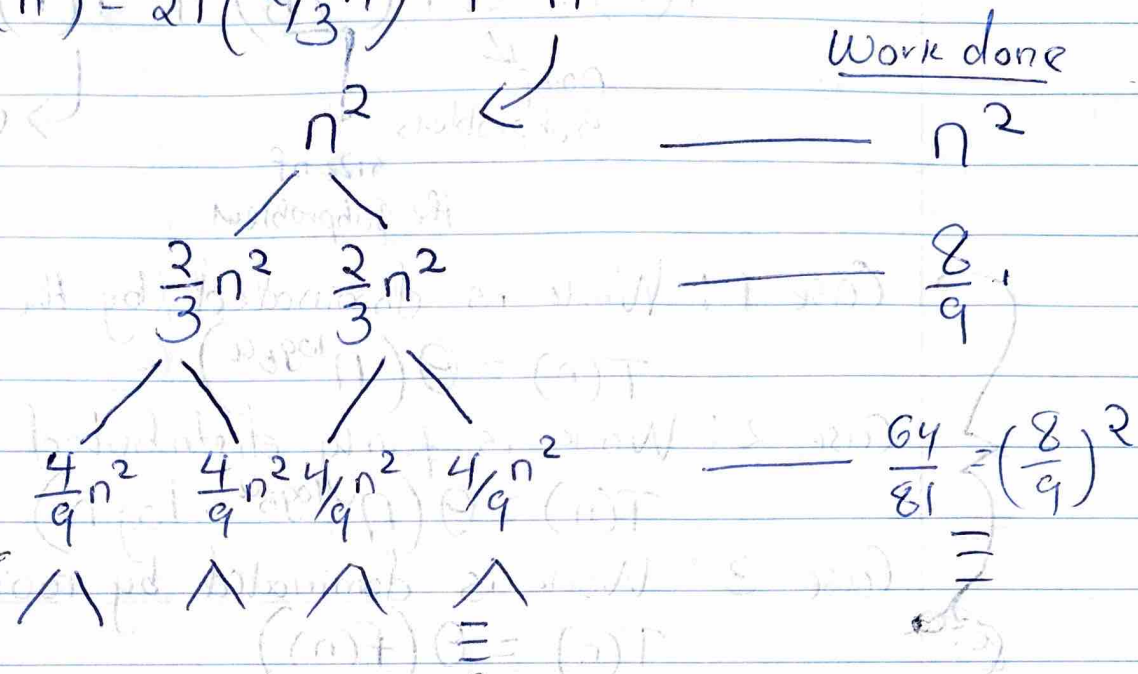
# Recurrences

① By using recursion trees.

$T(n) = 2T(2/3\, n) + n^2$




Case 1: Work decreases (root-heavy) - the cost per level shrinks $(r<1) \longrightarrow a \cdot \frac{1}{1-r}$

Case 2: Work is equal at each level (balanced

$r = 1 \longrightarrow 1^{st}$ term $\times$ no. of levels

e.g $n \times (\log_n 2 + 1) = \Theta(n \log n)$

Case 3: Work increases (leaf-heavy) - the cost per level increases $(r>1) \longrightarrow a \cdot \frac{r^2-1}{r-1}$

$$= n^2\left(1 + \left(\frac{8}{9}\right) + \left(\frac{8}{9}\right)^2 + \left(\frac{8}{9}\right)^3 + \cdots\right)$$

$$a \cdot \frac{1}{1-r} = n^2 \cdot \frac{1}{1 - 8/9} = 9n^2$$

$$= \Theta(n^2)$$

## (2) Master's Method (Does not work for every recurrence)

$$T(n) = aT(n/b) + f(n)$$

$a$ → no. of subproblems

$b$ → size of the subproblem

$f(n)$ → cost of work

Case 1: Work is dominated by the leaves

$$T(n) = \Theta(n^{\log_b a})$$

Case 2: Work is fairly distributed

$$T(n) = \Theta(n^{\log_b a} \cdot \log n)$$

Case 3: Work is dominated by root $(f(n))$

$$T(n) = \Theta(f(n))$$

e.g $T(n) = 2T(\tfrac{2}{3}n) + n^2$

$a = 2$, $b = \tfrac{2}{3}$, $f(n) = n^2$

$$\log_b a = \log_{2/3} 2$$

$$= \frac{\log 2}{\log 2/3}$$

$$= \frac{0.693}{0.401}$$

$$\approx 1.51 = n^{1.51}$$

$f(n) = n^2$

$\therefore n^{1.51} < n^2$

$\therefore$ Case 3

$$\therefore \Theta(f(n))$$

$$= \Theta(n^2)$$